Emmanouil Karantoumanis

+30 6981332554 | e.karantoumanis@uowm.gr / ekar.manolis@gmail.com | ekarantoumanis
Kozani, Greece | 20/05/1996

RESEARCH SUMMARY

Postdoctoral Researcher with extensive experience in the design, analysis, and optimization of algorithms, Mathematical Programming, and Artificial Intelligence (Machine Learning/LLMs). My research focuses on developing innovative methodologies for black-box optimization problems, derivative-free optimization, and adaptive sampling using surrogate models. I have a strong background in software development (Python, C++, MATLAB), infrastructure management, and the implementation of digital solutions across numerous research and development projects.

CURRENT STATUS

Postdoctoral Researcher 2026 – Present
Dept. of Electrical & Computer Engineering, University of Western Macedonia

- Research interests: **Optimization, Black-box optimization, Mathematical Programming, Algorithms, Machine Learning, Artificial Intelligence, Derivative-free optimization, Surrogate modeling, Adaptive sampling**

EDUCATION

Doctor of Philosophy (Ph.D.) 2019 – 2026
Dept. of Electrical & Computer Engineering, University of Western Macedonia

- Thesis Title: **Optimization algorithms for black-box problems**
- Research focus: Black-box optimization, Derivative-free optimization, Surrogate modeling, Adaptive sampling, Machine Learning

Diploma / Integrated Master 2014 – 2019
Dept. of Informatics & Telecommunications Engineering, University of Western Macedonia

- Grade: 7.04/10
- Thesis: **Energy consumption estimation in cloud computing** (Grade: 10/10)

PROFESSIONAL EXPERIENCE

Researcher - Project: ”Digital Innovation Hub SYNERGiNN EDIH” Sept. 2024 – Sept. 2026
University of Western Macedonia

- Design and implementation of digital services for businesses and public organizations using artificial intelligence methods.
- Application and specialization of digital services in the energy and environment sectors.

Researcher - Project: ”Digital Transformation of the Region of Western Macedonia” Sept. 2025 – Dec. 2025
University of Western Macedonia

- Implementation of artificial intelligence algorithms within the national program for the digital transformation of the Region of Western Macedonia.

Researcher - Project: ”Integration of AI Tools for the Regional Development Fund of W.M.” Sept. 2025 – Dec. 2025
University of Western Macedonia

- Implementation of artificial intelligence algorithms to support entrepreneurship and employment applications.

External Collaborator - Project: ”TOTEM” (Erasmus+) Nov. 2023 – Oct. 2025
Chamber of Kozani

- Creation and configuration of the Moodle e-learning platform to support the project’s educational objectives.

Researcher - Project: ”Adaptive Sampling in Electromagnetic Problems” Aug. 2025 – Aug. 2026
University of Western Macedonia

– Development and application of state-of-the-art derivative-free optimization methods to accelerate computations in smart telecommunication surfaces (6G wireless systems).

Researcher - Project: "ANALYST" Jun. 2024 – May 2025
University of Western Macedonia
– Development of a Moodle platform for training in cybersecurity and data analysis.

Researcher - Project: "Interconnected Smart Cities (Greece 2.0)" Nov. 2023 – Sept. 2025
University of Western Macedonia
– Design and implementation of a web integration platform for smart city vertical solutions.
– Development of an optimal waste collection routing application integrating Machine Learning techniques.

Researcher - Project: "Preparation of SYNERGiNN EDIH Proposal Submission" Jul. 2023 – Aug. 2024
University of Western Macedonia
– Writing and preparation of the proposal submission dossier for the SYNERGiNN EDIH project.

Researcher - Project: "Cost Optimization Algorithms in Cloud Services" Jun. 2023 – Aug. 2023
University of Western Macedonia
– Implementation and configuration of optimization algorithms aimed at reducing data transfer costs.

Researcher - Project: "Pilot Smart City Digital Applications for the Municipality of Kozani" Sept. 2022 – Oct. 2022
University of Western Macedonia
– Design and implementation of Machine Learning algorithms for the city's digital services needs.

Researcher - Project: "Cost Optimization Algorithms in Cloud Services" Mar. 2022 – May 2022
University of Western Macedonia
– Implementation and configuration of optimization algorithms to minimize data transfer costs.

Researcher - Project: "AGROTOUR" Jul. 2021 – Sept. 2023
University of Western Macedonia
– Development of object detection algorithms for early prediction of plant diseases (smart agriculture).

Researcher - Project: "Pilot Smart City Digital Applications for the Municipality of Kozani" Apr. 2021 – Sept. 2021
University of Western Macedonia
– Implementation of a network and specialized applications within the pilot smart city actions.

Researcher - Project: "Pilot Smart City Digital Applications for the Municipality of Kozani" Jan. 2021 – Mar. 2021
University of Western Macedonia
– Implementation of a network and specialized applications within the pilot smart city actions.

Researcher - Project: "Development of Advanced Scheduling Algorithms" Feb. 2020 – Jul. 2020
University of Western Macedonia
– Design, implementation, and application of optimization algorithms for black-box problems.
– Development of Python code to find solutions in optimization problems where the objective function and constraints are unknown.

PUBLICATIONS

International Journals

- **Karantoumanis, E.**, Zygiridis, T., & Ploskas, N. (2025). A Surrogate-Based Adaptive Sampling Approach for Electromagnetic Problems. *International Journal of Numerical Modelling: Electronic Networks, Devices and Fields*, 38(5), e70121, doi: 10.1002/jnm.70121.
- **Karantoumanis, E.**, & Ploskas, N. (2025). A surrogate-based adaptive sampling approach for mixed-integer black-box optimization problems. *Journal of Optimization Theory and Applications (JOGO)*, doi: 10.1007/s10898-025-01566-6.
- **Karantoumanis, E.**, Balafas, V., Louta, M., & Ploskas, N. (2024). Real-time plant disease detection from a small image dataset using data augmentation and deep learning methods. *Soft Computing (SOCO)*, doi: 10.1007/s00500-024-10348-3.
- **Karantoumanis, E.**, & Ploskas, N. (2024). Improving derivative-free optimization algorithms through an adaptive sampling procedure. *Results in Control and Optimization*, 100460, doi: 10.1016/j.rico.2024.100460.
- Balafas, V., **Karantoumanis, E.**, Louta, M., & Ploskas, N. (2023). Machine learning and deep learning for plant disease classification and detection. *IEEE Access*, doi: 10.1109/ACCESS.2023.3324722.

International Conferences & Proceedings

- **Karantoumanis, E.**, & Ploskas, N. (2026). A Constrained Grey-Box Optimization Method with Swarm-Guided Acquisition and Multi-Anchor Boundary Repair. *11th International Symposium & 33rd National Conference on Operational Research (HELORS 2026)*.
- **Karantoumanis, E.**, Mavromatidis, D., Mitropoulou, K., & Ploskas, N. (2026). Automated Job Matching using Large Language Models: A Case Study for the Region of Western Macedonia. In *29th Pan-Hellenic Conference on Progress in Computing and Informatics (PCI'2026)*, Association for Computing Machinery, New York, NY, USA, pp. 293–298, doi: 10.1145/3829216.3829312.
- Iosif, P., **Karantoumanis, E.**, Sarigiannidis, P., & Ploskas, N. (2026). Optimizing Waste Collection Routes Using IoT Sensors, Predictive Simulation, and VRP Techniques: A Case Study in Kozani, Greece. In *29th Pan-Hellenic Conference on Progress in Computing and Informatics (PCI'2026)*, Association for Computing Machinery, New York, NY, USA, doi: 10.1145/3829216.3829275.
- **Karantoumanis, E.**, & Ploskas, N. (2024). Optimizing concentrated solar power plants using adaptive sampling-based derivative-free algorithms. In *28th Pan-Hellenic Conference on Informatics (PCI 2024)*, Association for Computing Machinery, New York, NY, USA, doi: 10.1145/3716554.3716611.
- Louta, M., Papathanasiou, F., Damos, P., Ploskas, N., Dasygenis, M., Kyriakidis, T., Dimokas, N., Balafas, V., Chatzisavvas, A., Karampelia, I., **Karantoumanis, E.**, Mantas, N., Lazaridis, V., Vandikas, I., Stergiou, K., Angelidis, P., & Tsikos, E. (2022). Intelligent Pesticide and Irrigation Management in Precision Agriculture: The case of VELOS Project. *Proceedings of the 10th International Conference on Information and Communication Technologies in Agriculture, Food and Environment (HAICTA 2022)*, Athens, Greece. [CEUR Workshop Proceedings]
- **Karantoumanis, E.**, Balafas, V., Louta, M., & Ploskas, N. (2022). Computational comparison of image preprocessing techniques for plant diseases detection. In *2022 7th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)*, IEEE, pp. 1-5, doi: 10.1109/SEEDA-CECNSM57760.2022.9932972.
- **Karantoumanis, E.**, Krallis, X., & Ploskas, N. (2021). An experimental evaluation of sensors on a LoRaWAN-based GPS tracking system. In *25th Pan-Hellenic Conference on Informatics (PCI 2021)*, Association for Computing Machinery, New York, NY, USA, pp. 224–228, doi: 10.1145/3503823.3503865.
- **Karantoumanis, E.**, & Ploskas, N. (2021). A computational comparison of mixed-integer derivative-free optimization solvers in an adaptive sampling surrogate modeling framework. In *Book of Abstracts of the 31st European Conference on Operational Research (EURO 2021)*, Athens, Greece.
- **Karantoumanis, E.**, & Ploskas, N. (2020). Power consumption estimation in data centers using machine learning techniques. In: Kotsireas I., Pardalos P. (eds) *Learning and Intelligent Optimization. LION 2020. Lecture Notes in Computer Science*, vol 12096. Springer, Cham, doi: 10.1007/978-3-030-53552-0_20.
- **Karantoumanis, E.**, & Ploskas, N. (2020). A surrogate modeling and adaptive sampling framework for black-box optimization problems. *Virtual INFORMS Annual Meeting 2020*.

TEACHING EXPERIENCE

Teaching Assistant / Co-teaching in Postgraduate Program 2024 – 2026
University of Western Macedonia

– Teaching (2 course cycles) in the MSc "Digital Game Development" for the course: **Algorithmic Thinking**.

Teaching Assistant (Undergraduate) 2020 – 2025
University of Western Macedonia

– Algorithm Analysis and Design
– Discrete Mathematics
– Algorithms and Data Structures

SKILLS

Programming Languages: Python, C, C++, MATLAB, SQL, PHP, JavaScript, HTML, CSS

Artificial Intelligence & LLMs: Machine Learning & Deep Learning (PyTorch, TensorFlow, Scikit-Learn), LLMs (Hugging Face, Prompt Engineering, Fine-tuning, RAG)

DevOps & Sysadmin: Linux / Unix Administration, Docker Containers, Git (Version Control), CI/CD Pipelines, Shell Scripting (Bash), Cloud Infrastructure

Databases: Relational Databases (MySQL, PostgreSQL), NoSQL Databases, Database Design & Optimization

Foreign Languages: English (Proficiency - C2), German (Intermediate - B1), Greek (Native)